

Natural Language Processing

Lecture 4 Naïve Bayes & Smoothing

Part 1: Probability, Bayes' Theorem,
& Conditional Probability

Part 2: Worked Examples,
Zero-Frequency Problem,
& Laplace Smoothing

Part 3: Evaluation,
Summary, and
Future Directions

Learning Objectives

01 Explain the role of probability in NLP.

03 Understand the Naïve Bayes assumption.

05 Perform text classification manually.

07 Apply Laplace/add-one smoothing.

02 Apply Bayes' Theorem to classification.

04 Calculate class probabilities.

06 Explain the zero-frequency problem.

08 Implement Naïve Bayes using Python.

Why Probability in NLP?

SAMPLE INPUT

"Congratulations! You won a free prize."

HUMAN JUDGMENT

Experience and context tell us this is almost certainly a fraudulent message.

MACHINE ESTIMATION

$P(\text{Spam}) = 0.94$

$P(\text{Ham}) = 0.06$

DECISION: SPAM

Probability in Classification

CLASS 1 (C_1)

Spam

CLASS 2 (C_2)

Ham (Not Spam)

THE PREDICTION RULE

$$\text{Prediction} = \underset{c \in C}{\operatorname{argmax}} P(c \mid d)$$

For a new document d , we calculate the probability for each class. The class with the **largest probability** becomes the final prediction.

Bayes' Theorem

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

RESULT

Posterior

$P(A|B)$

LIKELIHOOD

Likelihood

$P(B|A)$

PRIOR

Prior

$P(A)$

EVIDENCE

Evidence

$P(B)$

Four Important Terms

PRIOR

Initial probability of a class before observing any evidence (baseline belief).

LIKELIHOOD

Probability of observing the evidence given that the class is true.

EVIDENCE

Total probability of observing the evidence across all possible classes.

POSTERIOR

Updated probability of the class after observing the evidence (final prediction).

Prior + Evidence → Posterior Probability

Prior Probabilities: A Simple Example

CORPUS STATISTICS

Ham Emails 70%

Spam Emails 30%

PRIOR CALCULATIONS

$P(\text{Ham}) = 0.70$

$P(\text{Spam}) = 0.30$

Definition: These represent our baseline knowledge or "initial belief" before looking at any specific words in a new email.

Conditional Probability in NLP

"What is the probability of A given that B has occurred?"

$$P(A \mid B)$$

NLP APPLICATION

$P(\text{Spam} \mid \text{"free"})$

MEANING

The probability that an email is **Spam**, given that it contains the specific word **"free"**.

From Bayes to Naïve Bayes

For a document containing words: $w_1, w_2, w_3, \dots w_n$

$$P(\text{Class} \mid \text{Words})$$

$\propto$

$$P(\text{Class}) \times P(\text{Words} \mid \text{Class})$$

Note: We focus on the numerator (Prior $\times$ Likelihood) because the denominator (Evidence) is constant across all classes during comparison.

The "Naïve" Assumption

$$P(w_1, w_2 \mid C) \approx P(w_1 \mid C) \times P(w_2 \mid C)$$

CORE CONCEPT

The model assumes that all features (words) are **conditionally independent** of each other, given the class label.

EXAMPLE: SPAM DETECTION

In "free prize winner", the words "**free**" and "**prize**" are treated as independent once we know the email is **Spam**.

Why Is It Called "Naïve"?

THE VIOLATION

DEPENDENCY EXAMPLE

"New York"

The word "York" is highly dependent on "New".
Treating them as independent is factually incorrect.

DEPENDENCY EXAMPLE

"Machine Learning"

Natural language is full of semantic and syntactic relationships that violate independence.

THE REALITY

Despite the "naïve" assumption, the model performs remarkably well.

Why? Because classification does not require a perfect language model. It only requires a **good decision boundary** to separate classes like Spam and Ham.

"All models are wrong, but some are useful." – George Box

Naïve Bayes Classification Steps

$$P(C | d) \propto P(C) \times \prod_i P(w_i | C)$$

01 CALCULATE PRIOR: Determine the frequency of each class in the training data.

02 CALCULATE LIKELIHOODS: Compute the probability of each word occurring within each class.

03 MULTIPLY: Combine the Prior with the product of all word Likelihoods for each class.

04 COMPARE: Evaluate the resulting scores across all possible classes.

05 SELECT: Predict the class that yields the highest overall score.

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Part 2: Worked Examples,
Zero-Frequency Problem,
& Laplace Smoothing

Step 1: Calculate Priors

EMAIL CONTENT

"free money"

"free prize"

"meeting today"

"project meeting"

CLASS

Spam

Spam

Ham

Ham

PRIOR CALCULATIONS

$$P(\text{Spam}) = 2/4 = 0.5$$

$$P(\text{Ham}) = 2/4 = 0.5$$

Step 2: Word Probabilities

FOR SPAM

"free"

PRESENT

"money"

PRESENT

FOR HAM

"free"

ABSENT

"money"

ABSENT

The Mathematical Conflict: The absence of "free" and "money" in the Ham training data will result in a probability of zero, which creates a significant problem for the classifier.

The Zero-Frequency Problem

Suppose: $P(\text{money} \mid \text{Ham}) = 0$ (Word never seen in Ham class)

$$\begin{aligned} & P(\text{Ham} \mid \text{"free money"}) \\ & \propto P(\text{Ham}) \times P(\text{free} \mid \text{Ham}) \times P(\text{money} \mid \text{Ham}) \\ & = 0.5 \times P(\text{free} \mid \text{Ham}) \times 0 = 0 \end{aligned}$$

CRITICAL FAILURE: A single unseen word can completely eliminate a class, regardless of all other evidence.

Why Zero Is Dangerous

$$0 \times \text{anything} = 0$$

EVIDENCE BIAS

A single unseen word can completely eliminate a class, even if all other words in the document strongly support it.

REAL-WORLD IMPACT

Training data is never exhaustive. New words are inevitable in production, making raw probabilities unstable.

THE SOLUTION: WE NEED SMOOTHING.

Laplace Smoothing (Add-One)

$$P(w|C) = \frac{\text{count}(w,C) + 1}{\text{total words in } C + |V|}$$

NUMERATOR

Ensures the count is **at least 1**, preventing zero probabilities.

DENOMINATOR

Adjusted by $|V|$ to ensure probabilities still sum to 1.

$|V|$

Represents the **Vocabulary Size** (total unique words in corpus).

The Logic of Adding One

$$0 + 1 \rightarrow 1$$

NON-ZERO RESULT

Ensures the numerator is always at least 1, preventing the entire product from collapsing to zero.

NUMERICAL STABILITY

Provides a small "safety" probability for terms that were not present in the training set.

UNIFORM PRIOR

Effectively assumes a uniform distribution over the vocabulary before any data is seen.

"I haven't seen this word in this class yet, but I won't assume it is impossible."

Worked Smoothing Example

INPUT PARAMETERS

Ham Content "meeting", "today",
"project"

Total Words (n) 3

Vocabulary Size (|V|) 5

CALCULATION FOR UNSEEN WORD

$$P(\text{word} \mid \text{Ham}) = \frac{0 + 1}{3 + 5} = 1/8$$

$$P(\text{word} \mid \text{Ham}) = 0.125$$

Multinomial Naïve Bayes

Unlike Bernoulli Naïve Bayes (which only checks presence/absence), the **Multinomial** variant accounts for **word frequencies** (counts).

INPUT DOCUMENT

"free free prize"

FEATURE EXTRACTION

free	2
prize	1

The model incorporates these counts when computing the likelihood. A word appearing multiple times provides stronger evidence for a class than a single occurrence.

Why Multinomial Naïve Bayes?

KEY ADVANTAGES

- ✓ **Efficiency:** Extremely fast and computationally lightweight.
- ✓ **Sparse Data:** Works exceptionally well with high-dimensional text vectors.
- ✓ **Small Datasets:** Effective even with limited training data.
- ✓ **Strong Baseline:** The standard starting point for text classification.

COMMON APPLICATIONS

Spam Detection

Sentiment Analysis

Topic Classification

News Categorization

Naïve Bayes and Bag-of-Words

LECTURE 3 RECAP

Text → Tokenization → Vocabulary → BoW Vector

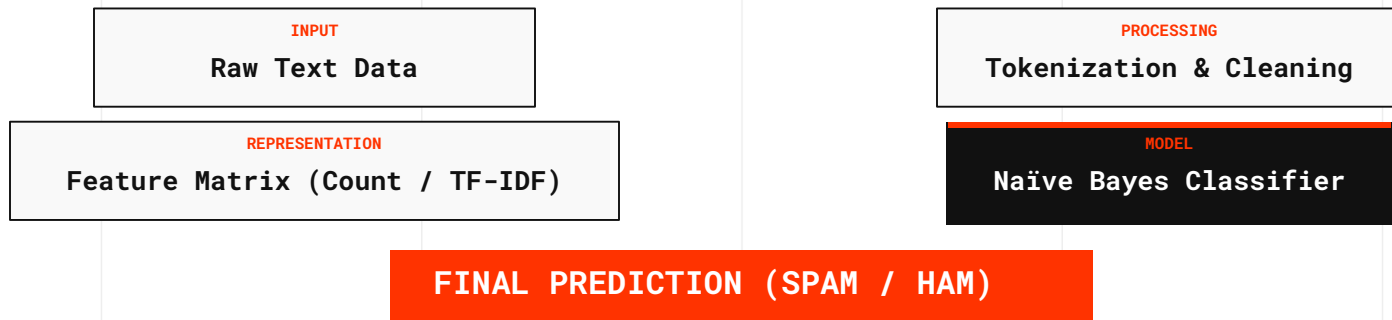


LECTURE 4 ADDITION

Feature Vector → Naïve Bayes → Class Prediction

Synergy: Representation (BoW) and Classification (Naïve Bayes) work together to transform raw, unstructured text into actionable insights and predictions.

Complete NLP Classification Pipeline



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The Confusion Matrix

	PREDICTED: SPAM	PREDICTED: HAM
ACTUAL: SPAM	TRUE POSITIVE TP	FALSE NEGATIVE FN
ACTUAL: HAM	FALSE POSITIVE FP	TRUE NEGATIVE TN

REAL-WORLD COST

In Spam Detection, a **False Positive** (Ham labeled as Spam) is often more costly than a **False Negative**, as it means an important email was missed.

Summary of Lecture 4

BAYES' THEOREM

The **mathematical foundation** for updating beliefs and calculating class probabilities from document evidence.

ZERO-FREQUENCY PROBLEM

The **catastrophic failure** where a single unseen word forces a zero probability across an entire class.

NAÏVE BAYES

An efficient classifier based on the **conditional independence** assumption between words.

LAPLACE SMOOTHING

The **add-one solution** that ensures numerical stability by preventing zero counts in likelihoods.

Quiz: Test Your Knowledge

01. Assumption

True or False: Naïve Bayes assumes words are independent given the class.

Think about the "Naïve" part of the name.

02. Calculation

If a word appears 0 times in a class of 10 total words, and $|V|=100$, what is the Laplace-smoothed probability?

Formula: $(\text{count} + 1) / (\text{total} + |V|)$

03. Application

Why is Multinomial Naïve Bayes preferred over Bernoulli for text classification?

Consider word frequencies vs. presence/absence.

04. Logic

Why do we ignore the "Evidence" term in the Bayes formula during classification?

How does it change across different classes?

Next Steps: Beyond Naïve Bayes

LECTURE 5

Logistic Regression

Moving from generative to discriminative models for more robust decision boundaries.

ADVANCED

SVMs & Forests

Exploring non-linear kernels and ensemble methods for complex classification tasks.

MODERN NLP

Neural Networks

Transitioning from Bag-of-Words to deep learning and contextual word embeddings.

GOAL: Understanding the trade-offs between simple baselines and complex models.

Conclusion & Q&A

Naïve Bayes remains one of the most powerful and efficient baselines in the NLP toolkit. Thank you for joining Lecture 4.

NEXT STEPS & RESOURCES

-  Course Portal: Lecture Notes
-  Worked Example Spreadsheet
-  Python Implementation Guide
-  Office Hours: Mon/Wed 2-4 PM